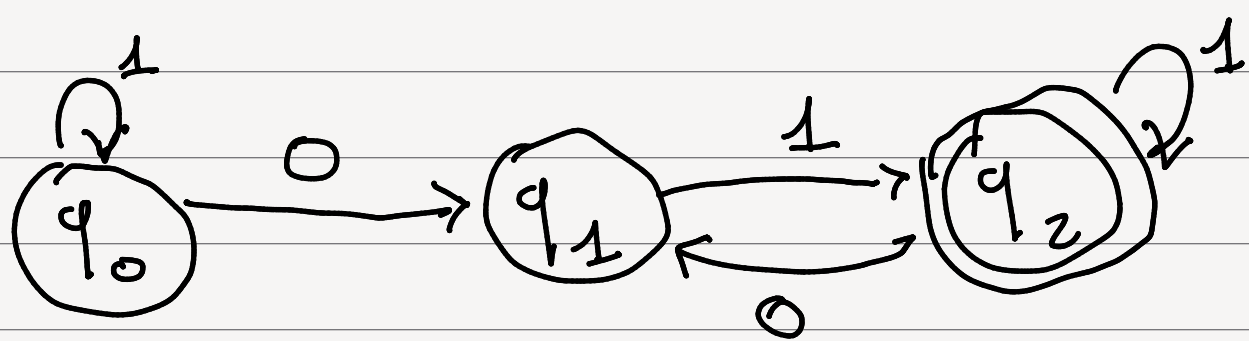
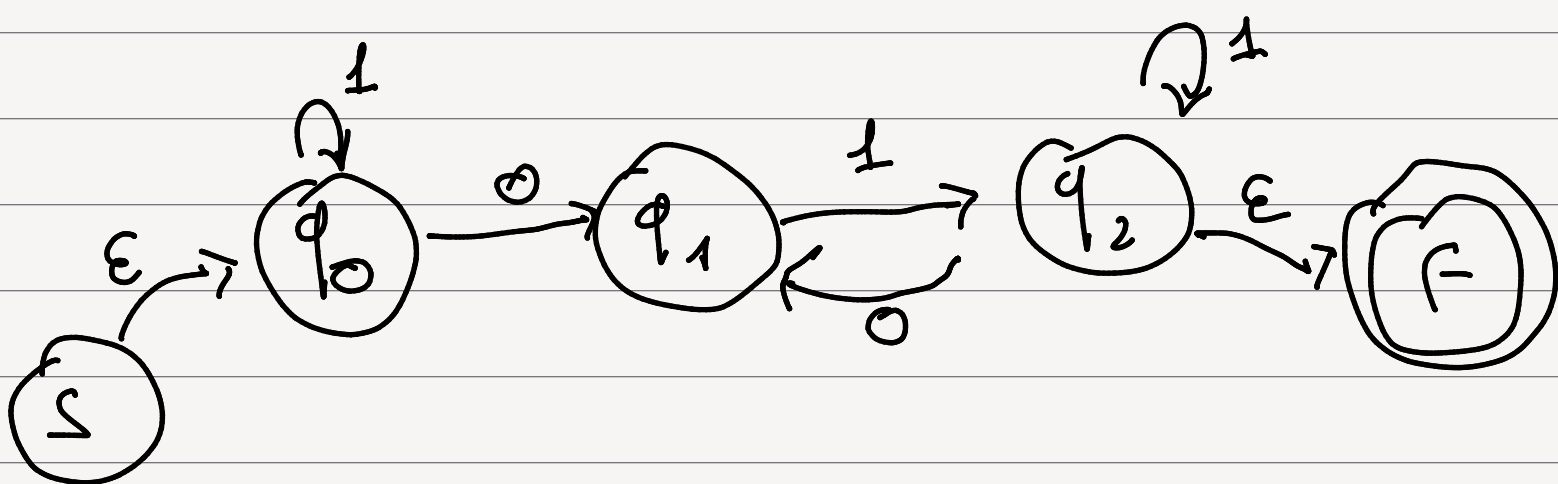


27. Stringhe binarie in cui ogni 0 ha almeno un 1 a destra



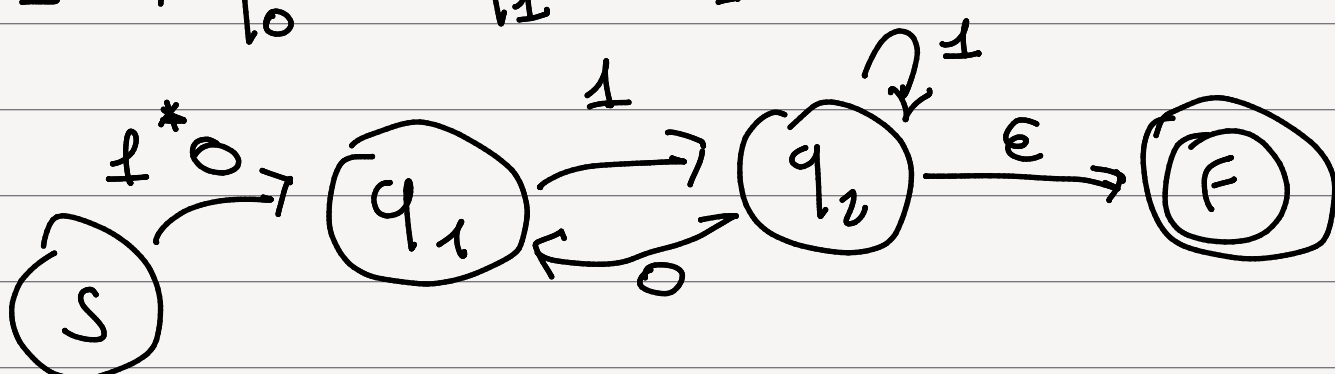
Trasformo in GNFA per la regex equivalente:



Elimino q_0 :

In	Out
S	q_1

$S \rightarrow q_0 \rightarrow q_1 = 1^* 0$



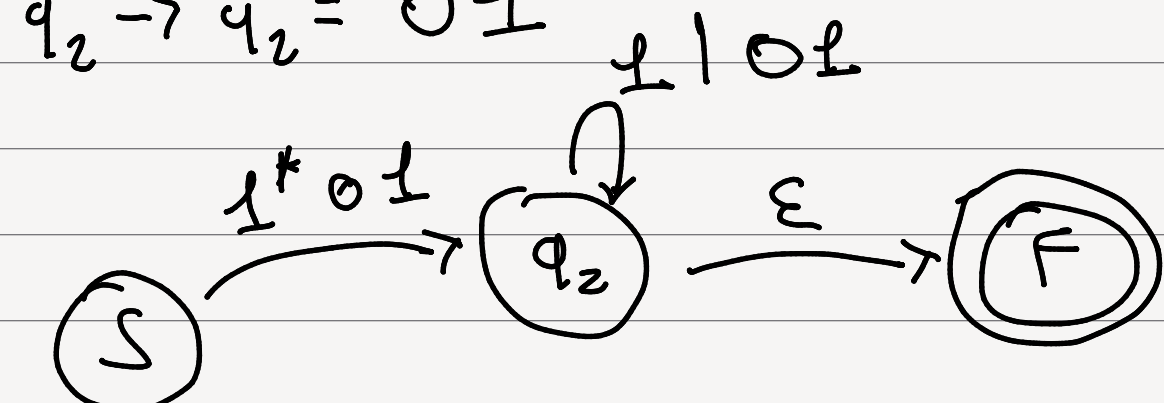
Elimino q_1 :

In	Out
S, q_2	q_2

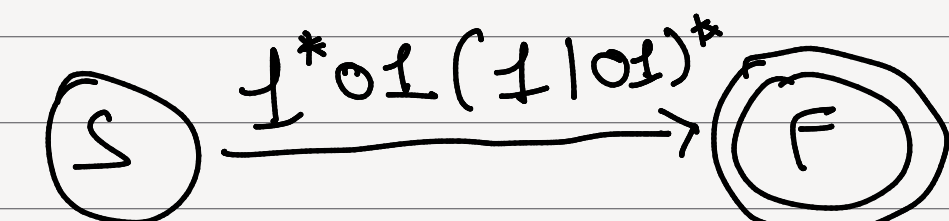
$S, q_2 \rightarrow q_2$

$S \rightarrow q_2 = 1^* 0 1$

$q_2 \rightarrow q_2 = 0 1$

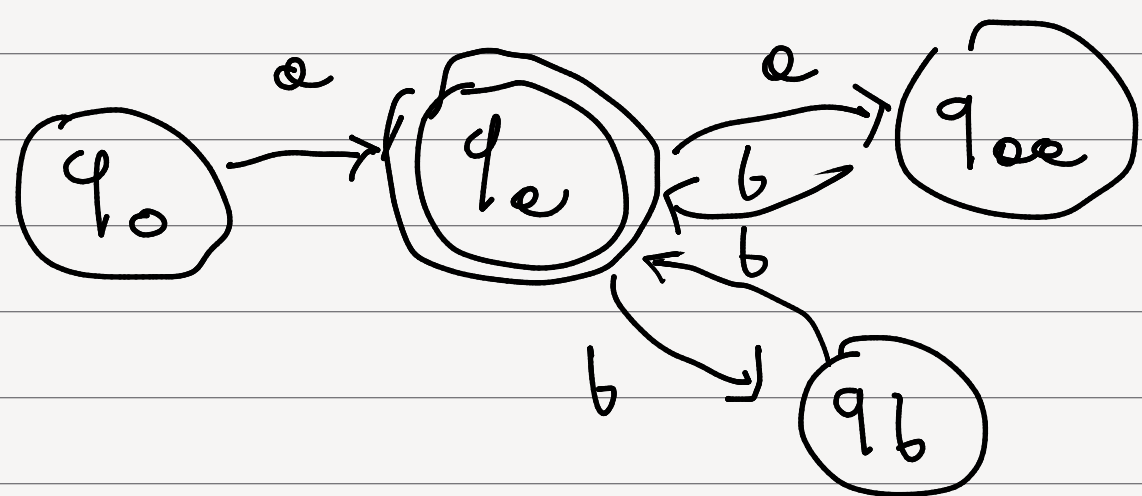


Elimino q_2 :

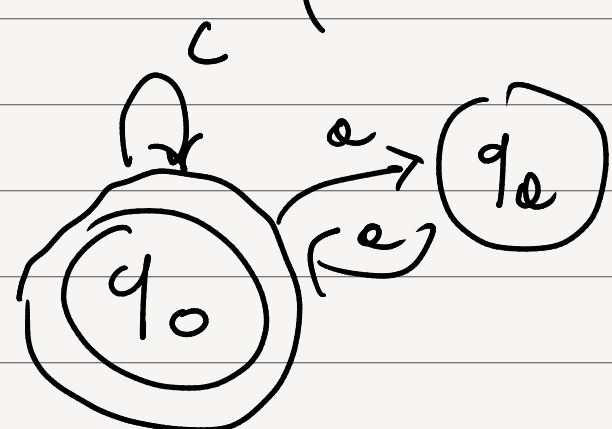


19. $a(ab|bb)^* \mid (aa|c)^*$

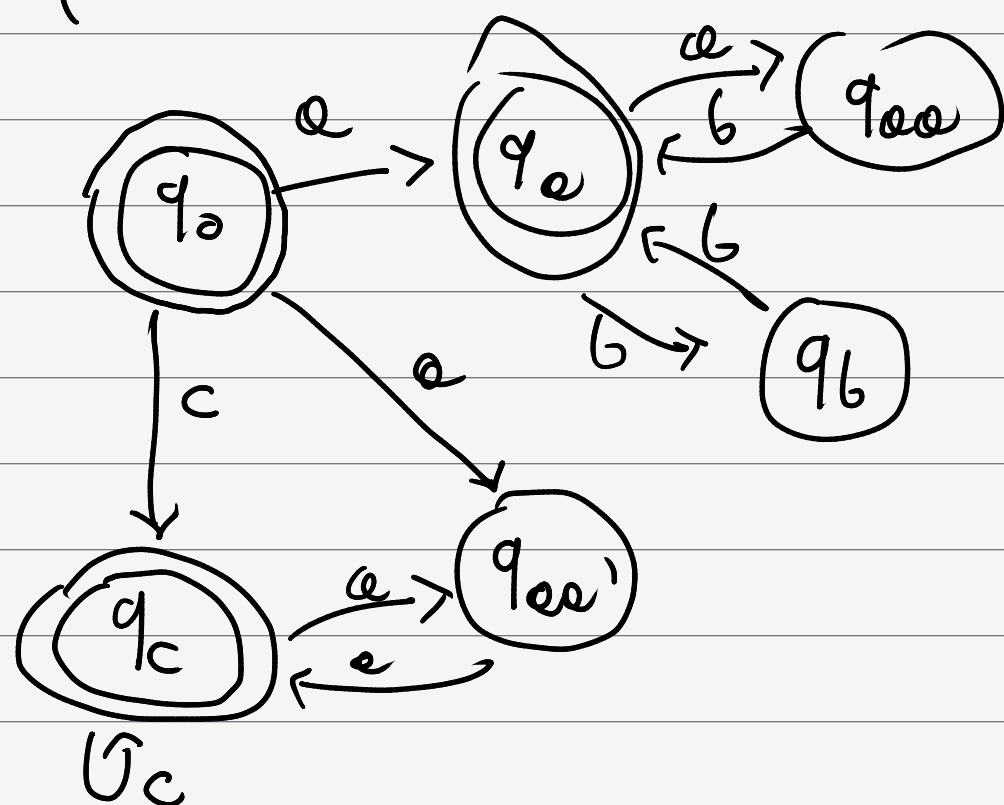
DFA per $a(ab|bb)^*$



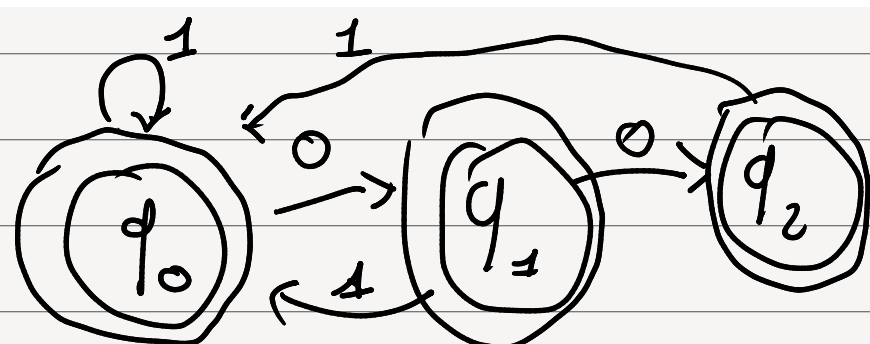
DFA per $(aa|c)^*$



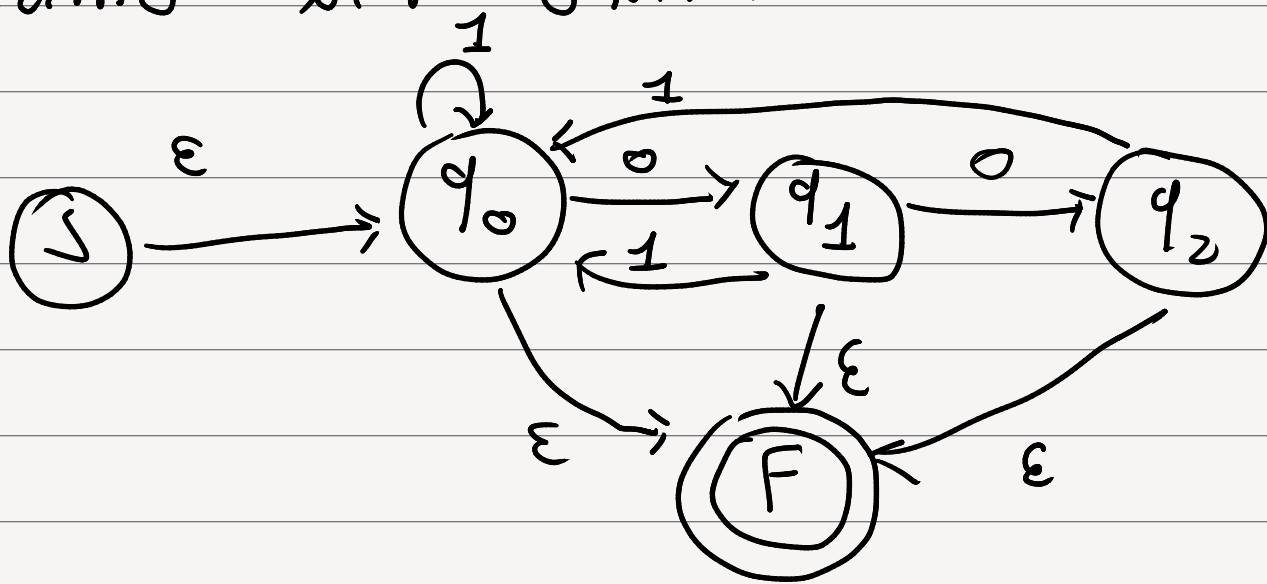
NFA per $a(ab|bb)^* \mid (aa|c)^*$



29. Stringhe binarie in cui non ci sono più di due 0 consecutivi



Transform to GNFA:

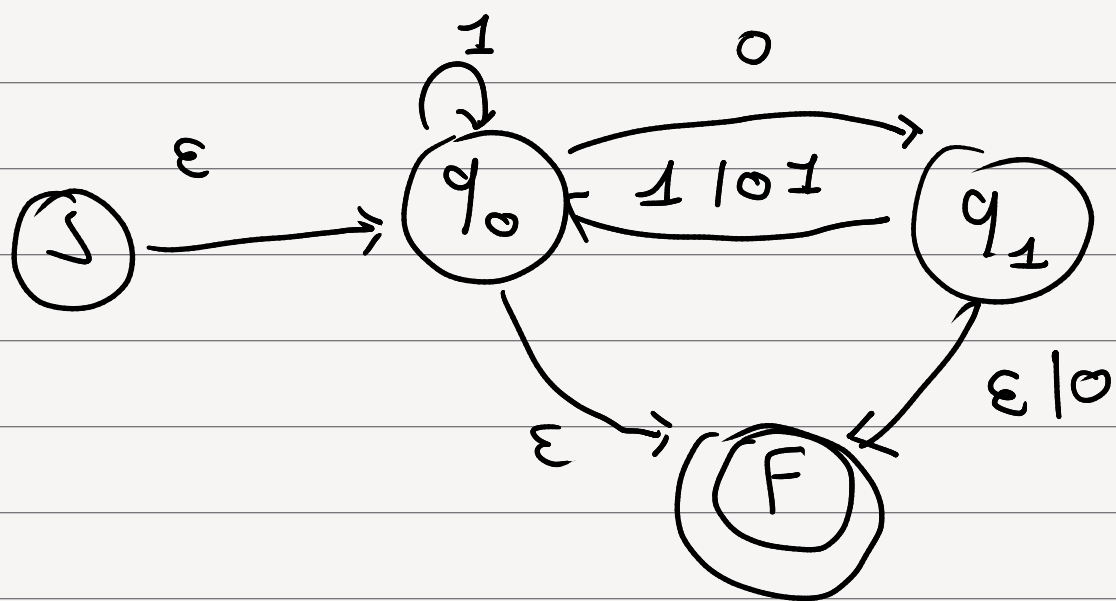


Path q_1 :

In Out
 q_1 q_0, F

$$q_1 \rightarrow q_0 = 01$$

$$q_1 \rightarrow F = 0$$



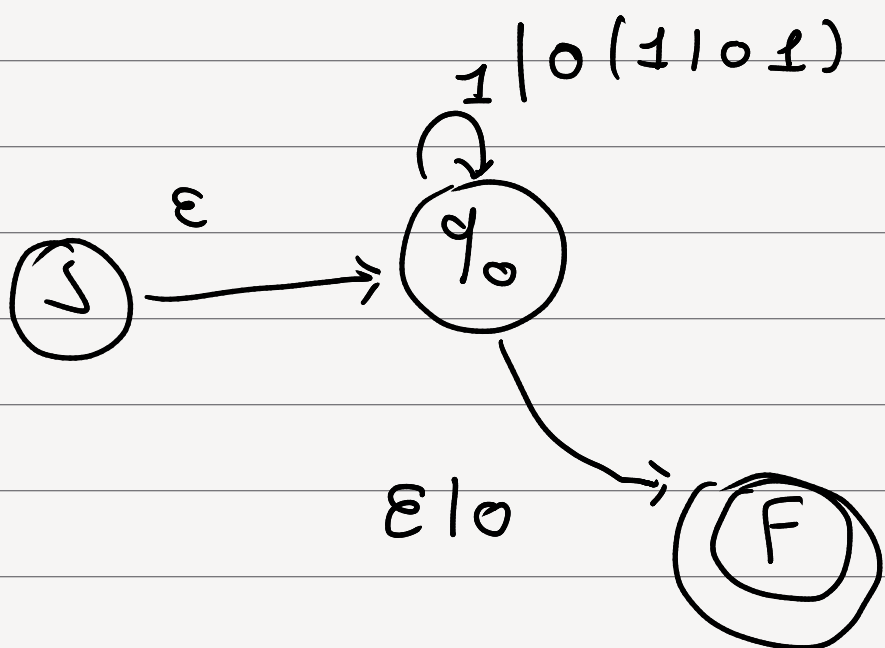
Path q_0 :

In Out

q_0 q_0, F

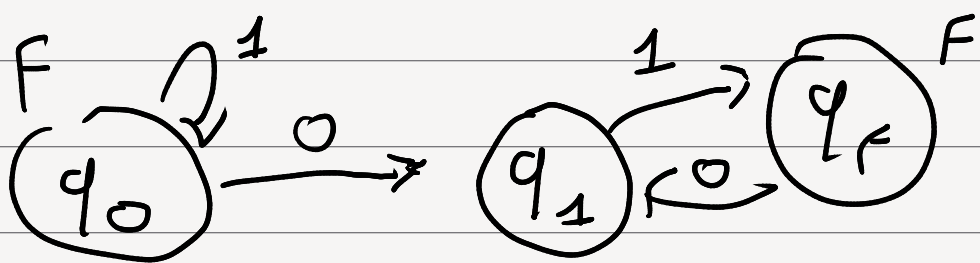
$$q_0 \rightarrow q_0 = 0(1|01)$$

$$q_0 \rightarrow F = \epsilon|0$$

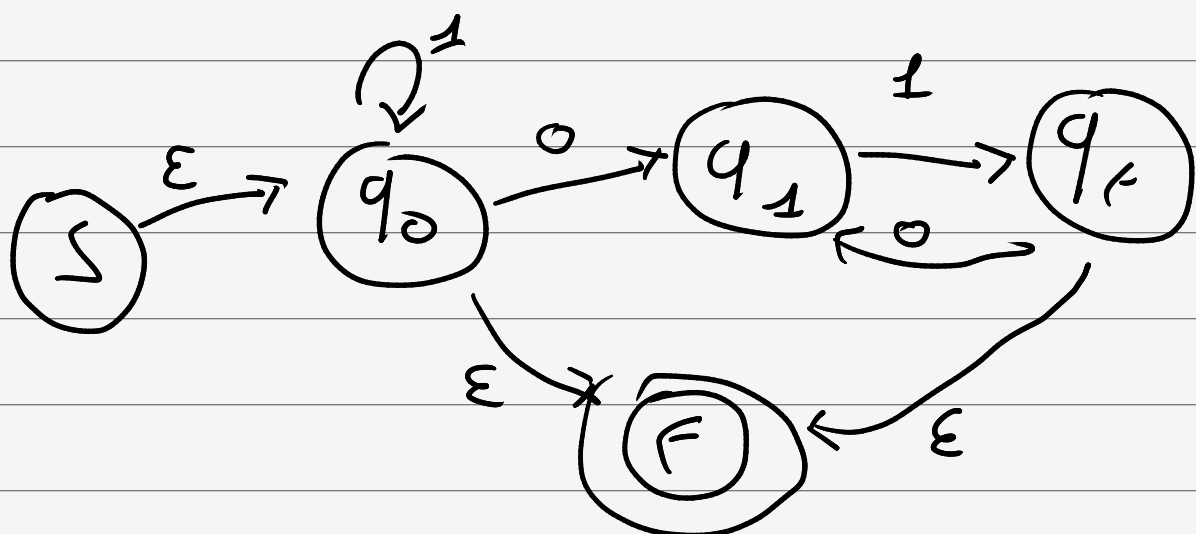


$$S \rightarrow F = (1|0(1|01))^* (\epsilon|0)$$

28. Stringhe binarie in cui ogni 0 ha esattamente un 1 a destra



GNFA:

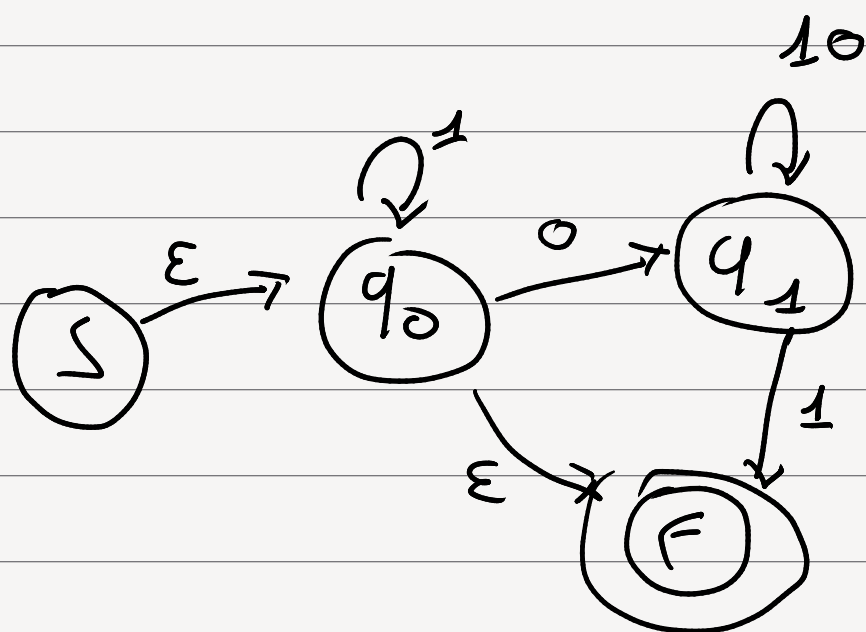


Tracce q_0 q_f :

In Out
 q_1 q_1, F

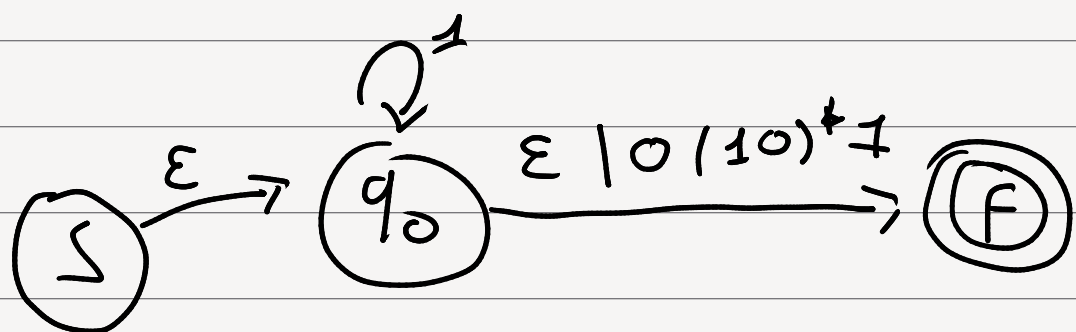
$q_1 \rightarrow q_1 = 10$

$q_1 \rightarrow F = 1$



Tracce q_1

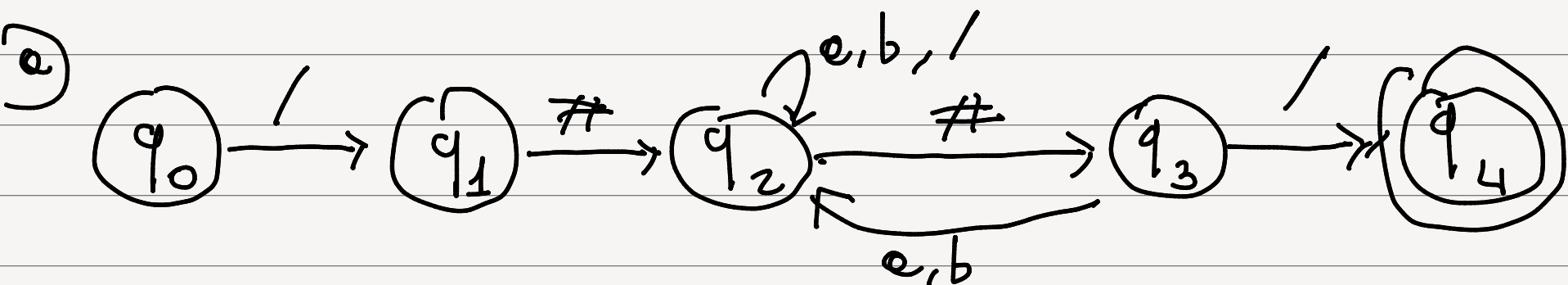
$q_0 \rightarrow F = \epsilon \mid 0(10)^*1$



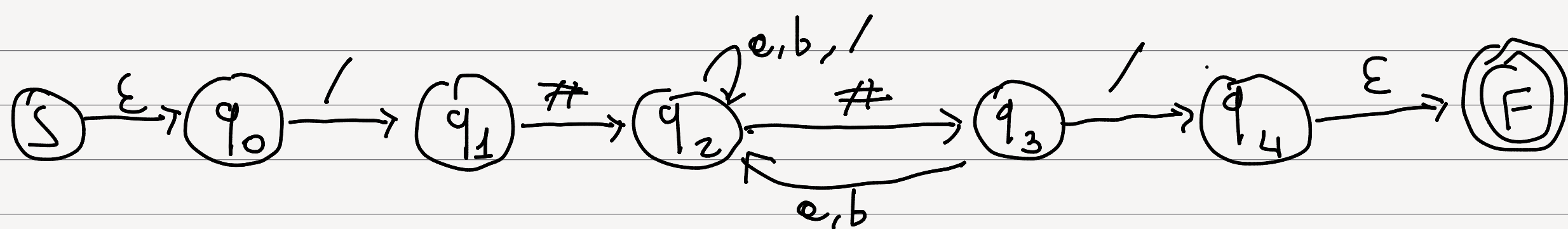
$S \rightarrow F = 1^* (\epsilon \mid 0(10)^*1)$

1.22 In alcuni linguaggi di programmazione, i commenti appaiono tra delimitatori come $/\#$ e $\#/\$. Sia C il linguaggio di tutte le stringhe di commento correttamente delimitate. Un elemento di C deve iniziare con $/\#$ e terminare con $\#/\$ ma non deve avere alcun $\#/\$ intermedio. Per semplicità, assumere che l'alfabeto per C sia $\Sigma = \{a, b, /, \#\}$.

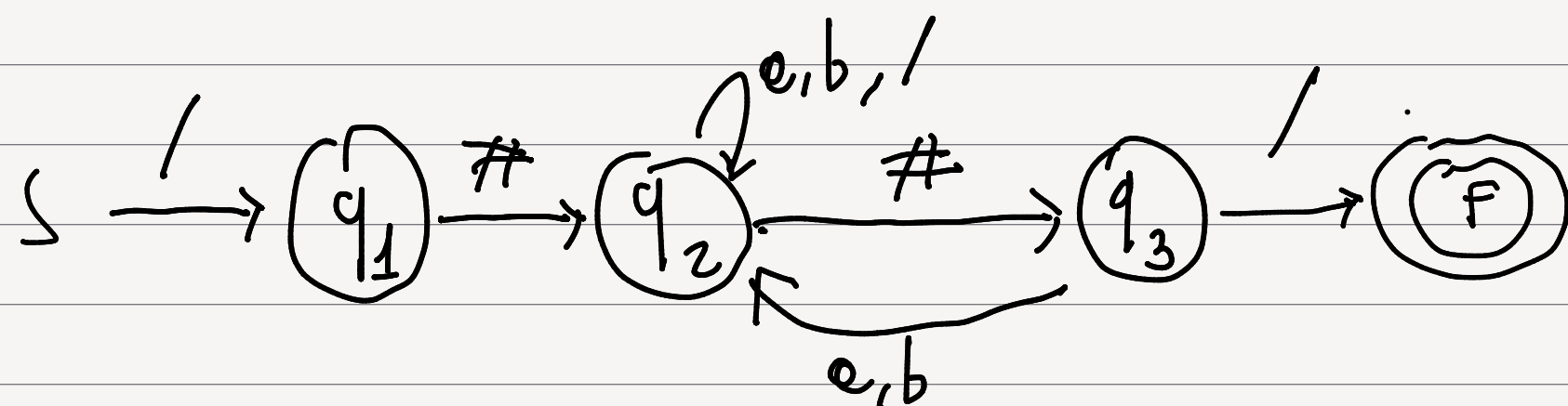
- Dare un DFA che riconosce C .
- Dare un'espressione regolare che genera C .



b) Costruiamo il GNFA che riconosce C .



che possiamo riscrivere come



Eliminiamo q_3 .

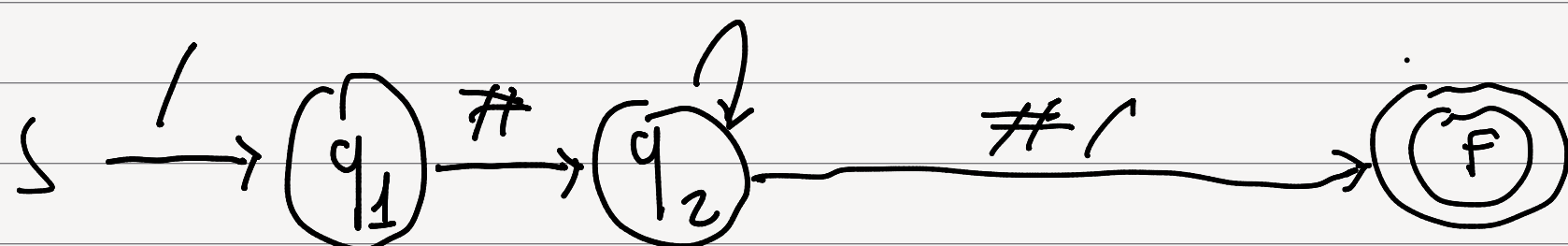
In Out

q_2 q_2, F

$q_2 \rightarrow q_2 = \#a \mid \#b$

$q_2 \rightarrow F = \#/\$

$(a \mid b \mid /)^* (\#a \mid \#b)$



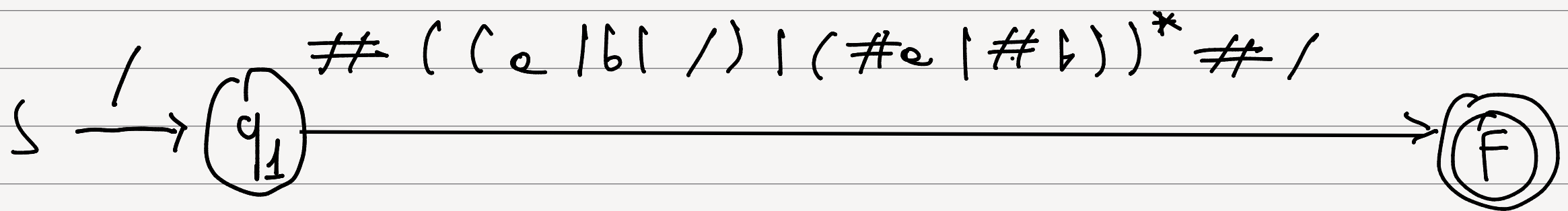
Eliminiamo q_2 :

In Out

q_1

F

$$q_1 \rightarrow F = \# ((a|b|/)|(\#a|\#b))^* \# /$$



che diventa:

